



Problem

Problem Statement

You are given a grid of dimensions $N \times M$. Each cell contains one of the following values:

0 $\rightarrow$ Empty cell 1 $\rightarrow$ Fresh orange 2 $\rightarrow$ Rotten orange

A rotten orange can rot an adjacent fresh orange in 1 unit of time.

A rotten orange at position (i, j) can rot fresh oranges at:

$(i-1, j) \rightarrow (i+1, j) \rightarrow (i, j-1) \rightarrow (i, j+1) \rightarrow$ Right

Find the minimum time required to rot all fresh oranges.

If it is impossible to rot all fresh oranges, print -1.

If there are no fresh oranges initially, the required time is 0.

Submissions

Leaderboard

Input Format

The first line contains an integer T , the number of test cases. For each test case: The first line contains two integers N and M , representing the number of rows and columns. The next N lines contain M space-separated integers representing the grid.

Constraints

$1 \leq T \leq 100$ $1 \leq N, M \leq 500$ $\text{grid}[i][j] \in \{0, 1, 2\}$

Discussions

Output Format

Print a single integer representing the minimum time required to rot all fresh oranges.

Print -1 if some fresh oranges can never become rotten

Sample Input 0

```
3 3
2 1 1
1 1 0
0 1 1
```

Sample Output 0

```
4
```

Sample Input 1

```
3 3
2 1 1
0 1 1
1 0 1
```

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Language Java 15

```
1  import java.util.*;
2
3  public class Main {
4
5      public static int orangesRotting(int[][] g
6          int n = grid.length;
7          int m = grid[0].length;
8
9          Queue<int[]> queue = new LinkedList<>()
10         int fresh = 0;
11         int time = 0;
12
13         // Find rotten and fresh oranges
14         for (int i = 0; i < n; i++) {
15             for (int j = 0; j < m; j++) {
16                 if (grid[i][j] == 2) {
17                     queue.add(new int[]{i, j})
18                 } else if (grid[i][j] == 1) {
19                     fresh++;
20                 }
21             }
22         }
23
24         int[][] directions = {
25             {-1, 0},
```

Line: 86 Col: 2



Upload Code as File

Run Code

Submit Code



Test against custom input

Congratulations

You solved this challenge. Would you like to challenge your friends?



Test case 0



Test case 1

Compiler Message

Success